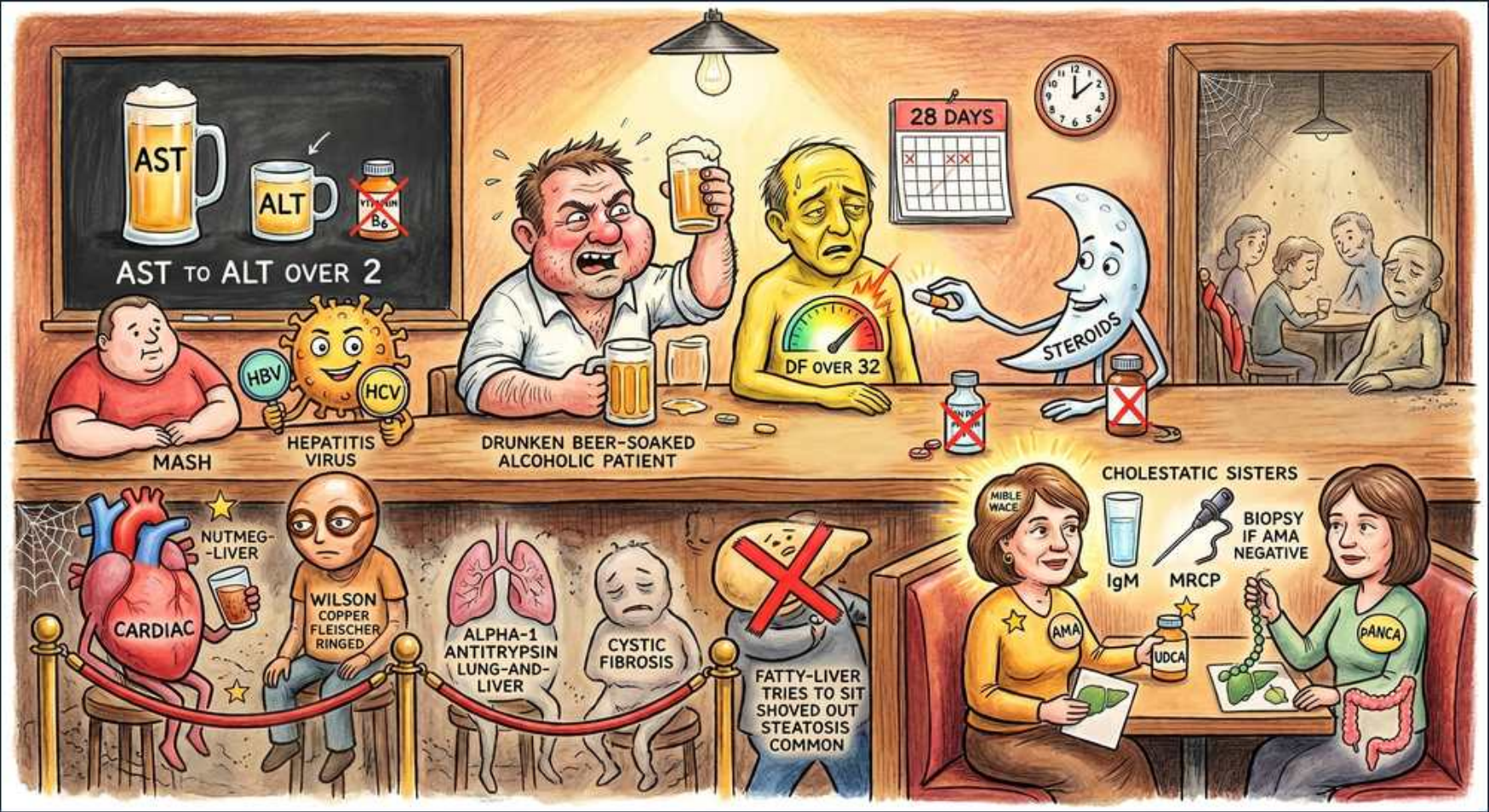


Cirrhosis — Etiologies



1. AST:ALT >2 (alcohol)

WHAT YOU SEE

A chalkboard 'AST TO ALT OVER 2' with beer mug, ALT bottle, and B6 vial, top-left.

WHAT IT ENCODES

Alcoholic liver disease classically shows AST:ALT ratio >2:1 with both usually <300. The relatively low ALT reflects pyridoxine (vitamin B6) deficiency needed for ALT synthesis. AST 'Scotch and Tonic' mnemonic. GGT and MCV are also elevated.

2. DF >32 → steroids

WHAT YOU SEE

A green gauge 'DF OVER 32' with a steroids bottle/moon figure, center.

WHAT IT ENCODES

Severe alcoholic hepatitis with Maddrey discriminant function >32 (or MELD high / encephalopathy) is treated with corticosteroids (prednisolone 40 mg/day). Reassess with the Lille score at day 7 — if Lille >0.45 (non-responder) stop steroids. DF uses PT and bilirubin.

3. Pentoxifylline/NAC (X)

WHAT YOU SEE

A steroids bottle marked with a red X, and distractor agents, center.

WHAT IT ENCODES

Pentoxifylline and N-acetylcysteine are NOT first-line for alcoholic hepatitis; the answer when DF>32 is corticosteroids. Pentoxifylline (a TNF inhibitor) showed no clear survival benefit in trials (STOPAH). NAC is adjunctive at best.

BOARD TRAP

"Pentoxifylline or NAC is first-line for severe alcoholic hepatitis" → wrong; give corticosteroids when Maddrey DF >32, then check Lille at day 7.

4. MASH/NASH

WHAT YOU SEE

A round person labeled 'MASH', bar stool, lower-left.

WHAT IT ENCODES

MASH/NASH (metabolic dysfunction-associated steatohepatitis) is now a leading cirrhosis cause, linked to obesity, T2DM, dyslipidemia. Diagnosis of exclusion with steatosis + inflammation. Manage with weight loss; resmetirom newly approved for MASH with fibrosis.

5. HBV / HCV

WHAT YOU SEE

Two virus mascots labeled 'HBV' and 'HCV', lower-left.

WHAT IT ENCODES

Chronic hepatitis B and C are major viral causes. HCV is now curable with direct-acting antivirals (>95%). HBV managed with entecavir/tenofovir. Both carry hepatocellular carcinoma risk requiring surveillance (US +/- AFP every 6 months in cirrhosis).

6. Cardiac (nutmeg) — rare (X)

WHAT YOU SEE

A heart labeled 'CARDIAC / NUTMEG LIVER', lower-left.

WHAT IT ENCODES

Cardiac cirrhosis ('nutmeg liver' from chronic hepatic congestion in right heart failure/constrictive pericarditis) is now RARE and should not be picked as a common cause. Modern HF management made it uncommon. SAAG would be high (>1.1) here.

BOARD TRAP

"Cardiac cirrhosis is a common etiology" → wrong; nutmeg-liver cardiac cirrhosis is now rare. Think alcohol, MASH, HBV/HCV first.

7. Wilson disease

WHAT YOU SEE

A figure 'WILSON COPPER FLEISCHER RINGED' (Kayser-Fleischer), lower-left.

WHAT IT ENCODES

Wilson disease: copper overload in a YOUNG patient (<40), LOW ceruloplasmin, high urinary copper, Kayser-Fleischer rings, neuropsychiatric signs, hemolytic anemia. Treat with chelation (penicillamine, trientine) or zinc. Screen first-degree relatives.

8. Alpha-1 antitrypsin

WHAT YOU SEE

Lungs-and-liver figure 'ALPHA-1 ANTITRYPSIN LUNG AND LIVER', lower-center.

WHAT IT ENCODES

Alpha-1-antitrypsin deficiency hits LUNG (panacinar/basal emphysema, esp. smokers) AND LIVER (PAS-positive diastase-resistant globules from misfolded protein accumulation). Check A1AT level and Pi typing (ZZ phenotype). No specific liver drug; transplant for end-stage.

9. Cystic fibrosis

WHAT YOU SEE

A figure labeled 'CYSTIC FIBROSIS', lower-center.

WHAT IT ENCODES

Cystic fibrosis can cause focal biliary cirrhosis from inspissated bile in CFTR dysfunction. Consider in younger patients with pulmonary/pancreatic disease. Ursodeoxycholic acid may be used; CFTR modulators changing the landscape.

10. Steatosis common (X note)

WHAT YOU SEE

A red X over 'FATTY-LIVER TRIES TO SIT — STEATOSIS COMMON', lower-center.

WHAT IT ENCODES

Simple steatosis is extremely common but is NOT itself cirrhosis; only steatohepatitis (MASH/NASH) with fibrosis progresses. Don't over-call bland fatty liver as a cirrhotic etiology in itself. Risk stratify fibrosis with FIB-4/elastography.

BOARD TRAP

"Plain steatosis = cirrhosis" → wrong; bland fatty liver is common and usually benign; only MASH with fibrosis progresses to cirrhosis.

11. PBC = AMA, UDCA

WHAT YOU SEE

Left 'cholestatic sister' with 'AMA', IgM, MRCP, 'UDCA' bottle, right side.

WHAT IT ENCODES

Primary biliary cholangitis: middle-aged woman, fatigue, PRURITUS, anti-MITOCHONDRIAL antibody (AMA) positive, high IgM, cholestatic pattern. Treat with ursodeoxycholic acid (UDCA); obeticholic acid second-line. If AMA-NEGATIVE, do a liver BIOPSY to confirm.

BOARD TRAP

"AMA-negative rules out PBC" → wrong; if clinical/biochemical picture fits but AMA is negative, obtain a liver biopsy to confirm AMA-negative PBC.

12. PSC = p-ANCA, beaded ducts

WHAT YOU SEE

Right 'cholestatic sister' labeled 'pANCA' with beaded-duct MRCP, right side.

WHAT IT ENCODES

Primary sclerosing cholangitis: young man, p-ANCA positive, MRCP shows 'beaded' multifocal stricturing of intra/extrahepatic ducts. Strongly associated with IBD (especially ulcerative colitis). Major cholangiocarcinoma risk (also colorectal cancer); no proven medical cure, transplant for end-stage.
